

# WP-MIP GFDL Vortex Tracker Outputs (Version 1.0)

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This dataset represents the second release (Version 1.0) of GFDL Vortex Tracker (Marchok 2021) outputs, which was made available via GitHub in July 2026 specifically for the WP-MIP. Comprehensive details regarding the dataset and its output formats are provided in the accompanying documentation, "GFDL Vortex Tracker Format Instructions for WP-MIP". The updates in this release version for individual models are included in the section of “**Additional Notes**” listed at the end of this document.

The model indices and contributing organizations are summarized in Table 1. As of July 2026, the Vortex Tracker has processed all accessible WP-MIP model forecasts, encompassing both the Core and SP1 datasets. The initialization frequency for each model, either daily or every three days, is represented by the color coding within the table cells.

| Identifier | Contributor | Forecasts OIC |        |          | Forecasts SIC |        |          |
|------------|-------------|---------------|--------|----------|---------------|--------|----------|
|            |             | AI            | Hybrid | Physical | AI            | Hybrid | Physical |
| babj       | CMA         | cmao          |        |          |               |        |          |
| cwao       | ECCC        | cwao          | cwho   | cwpo     | cwas          | cwhs   | cwps     |
| ecmf       | ECMWF       | ecao          | echo   | ecpo     |               |        |          |
| edzw       | DWD         | dwao          |        | dwpo     |               |        | dwps     |
| egrr       | UKMO        | ukao          |        | ukpo     |               |        | ukps     |
| gfdl       | GFDL        |               |        | shpo     |               |        | shps     |
| kwbc       | NOAA        | gfao          |        | gfpo     | gfas          |        | gfps     |
| rjtd       | JMA         | jmao          |        | jmpo     |               |        |          |
| rksl       | KMA/KIAPS   |               |        | kmpo     |               |        | kmps     |
| rums       | RAS         |               |        | rspo     |               |        | rsps     |
| sbsj       | CPTEC/INPE  |               |        | bzpo     |               |        | bzps     |

Table 1: Key to model indices for WP-MIP GFDL Vortex Tracker outputs. Cell coloring identifies initialization frequency: light green for daily, dark green for three-day intervals. Red indicates tracker runs are inhibited by missing essential variables. Grey cells denote instances where contributors did not provide forecast data.

Tropical cyclones are tracked using metrics based on the following variables provided by each model: sea-level pressure, u- and v-component winds at 10 m, 850 and 700 hPa, and geopotential heights at 850 and 700 hPa. The tracker also provides complete thermodynamic

diagnostics of Cyclone Phase Space (Hart 2003) and a check for a warm core in the 300-500 mb layer. These diagnostics are based on the geopotential heights at 925, 850, 700, 500, 400, and 300 hPa, as well as the temperature fields at 500, 400, and 300 hPa. The columns for these diagnostics will be left blank if a model does not provide all of the above mentioned variables.

### **Additional Notes listed by the data contribution centers:**

Note: The default model version used to produce the tracker outputs is “00”, e.g. pm\_00, ai\_00, or hy\_00. Exceptions are listed below for each center.

#### **1. ECCC:**

- The wind fields used to produce tracker outputs in the release version v0.0 for the three OIC models (cwao, cwho, and cwpo) were interpolated incorrectly. Therefore, the following three files in release version v0.0 were deleted:  
WP-MIP\_GFDL\_Vortex\_Tracker\_Outputs\_cwao\_v0.0.tar  
WP-MIP\_GFDL\_Vortex\_Tracker\_Outputs\_cwho\_v0.0.tar  
WP-MIP\_GFDL\_Vortex\_Tracker\_Outputs\_cwpo\_v0.0.tar
- The three OIC models (cwao, cwho, and cwpo) in this release (v1.0) are updated to daily data with correct wind fields.
  - cwao is based on “ai\_01”; cwho is based on “hy\_01”; cwpo is based on “pm\_01”

#### **2. ECMWF:**

- The hybrid (echo) and physical (ecpo) models are updated to daily datasets.
- The hybrid model (echo) in this release (v1.0) is based on the model version “hy\_01”, which is different from “hy\_00” used in the previous release (v0.0).

#### **3. DWD:**

- About the OIC/AI model (dwao):
  - The tracker used the model version “ai\_01”.
  - The tracker turned off using the wind fields at 700 and 850 hPa for tracking storms due to many missing data points (10%) in these 4 variables.
  - The thermodynamic diagnostics are not provided due to the lack of geopotential heights.

#### **4. UKMO:**

- The sea-level pressure values used to produce tracker outputs in release version v0.0 for the OIC/physical model (ukpo) were incorrect. The file:  
WP-MIP\_GFDL\_Vortex\_Tracker\_Outputs\_ukpo\_v0.0.tar  
in release version v0.0 was deleted.

- The updated OIC/physical (ukpo) tracker output in this release (v1.0) is based on the model version “pm\_01” with correct sea-level pressure field.

5. NOAA:

- The thermodynamic diagnostics missing in some forecast lead times in the previous release (v0.0) for the OIC/physical model (gfpo) are all available in this release (v1.0).

6. JMA:

- The thermodynamic diagnostics are not provided to OIC/AI (jmao) and OIC/physical (jppo) models due to the lack of required variables.

7. RAS:

- The tracker output for the OIC/physical model (rspo) is based on the model version “pm\_01” in this release (v1.0).

8. CPTEC/INPE:

- The tracker output for both OIC/physical (bzpo) and SIC/physical (bzps) models are based on the model version “pm\_01” in this release (v1.0).

References:

Hart, R. E., 2003: A Cyclone Phase Space Derived from Thermal Wind and Thermal Asymmetry. *Mon. Wea. Rev.*, **131**, 585–616,  
[https://doi.org/10.1175/1520-0493\(2003\)131<0585:ACPSDF>2.0.CO;2](https://doi.org/10.1175/1520-0493(2003)131<0585:ACPSDF>2.0.CO;2).

Marchok, T., 2021: Important factors in the tracking of tropical cyclones in operational models. *J. Appl. Meteor. Climatol.*, <https://doi.org/10.1175/JAMC-D-20-0175.1>.